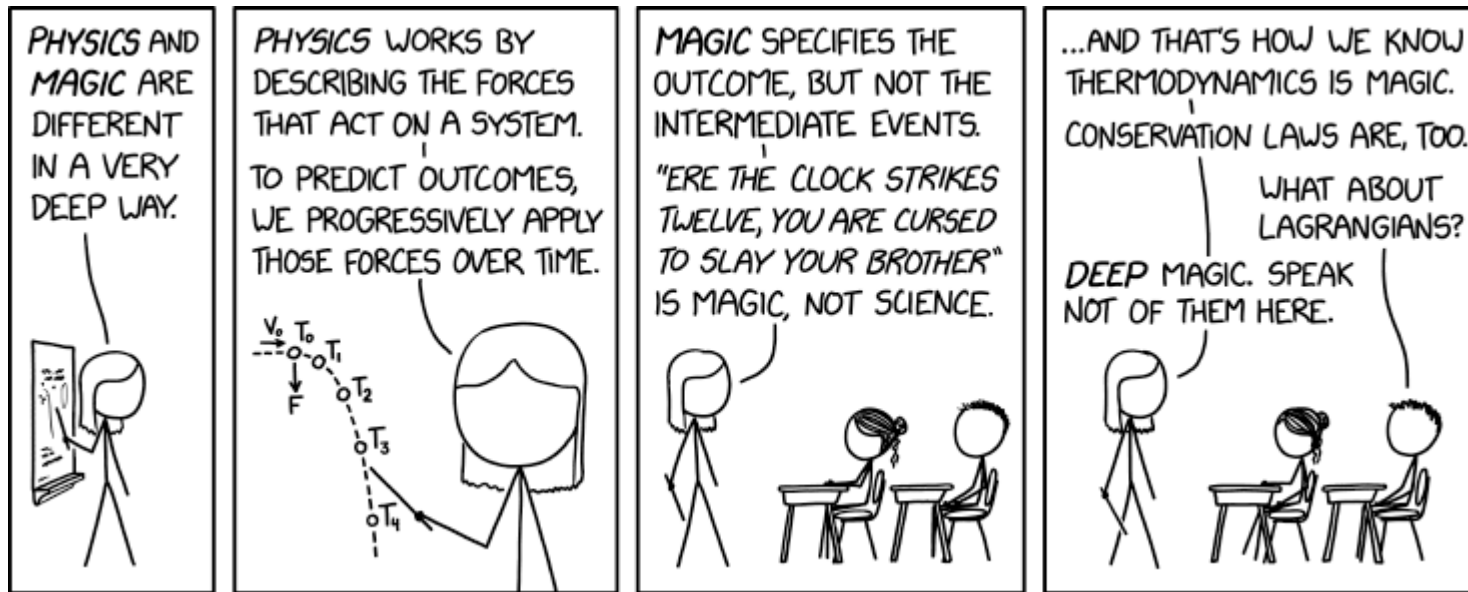




<https://xkcd.com/2904/>

## Classic Mechs EP5: Non-linear Dynamics and into the Lagrangian land

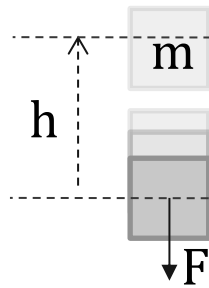
# Classic Mechs EP5.1: Non-linear dynamics

## Learning objectives

- 1) We will use non-linear potential field to study non-linear dynamics
- 2) We will use all the knowledge we acquired so far to linearize non-linear system at fixed point
- 3) Sometimes, we can have a very educated guess of the whole dynamic systems

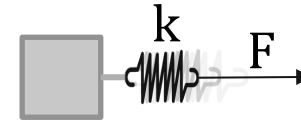
# Potential energy and conservative force

The force to drop a block due to the gravity:



$$F = -\frac{\partial(mgh)}{\partial h} = -mg$$

The force from a compressed spring:



$$F = -\frac{\partial(\frac{1}{2}kx^2)}{\partial x} = -kx$$

We all agree on this intuition (in 1D):  $F = -\frac{\partial U}{\partial x}$

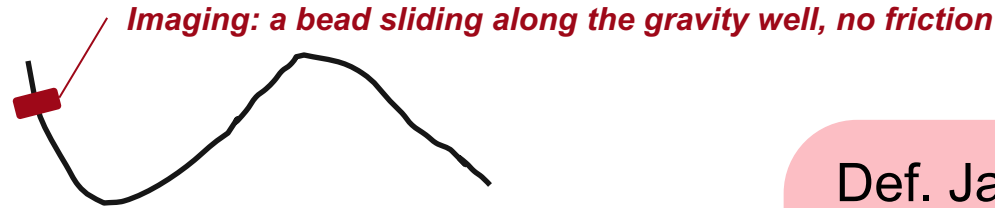
Expand it to 3D:  $\vec{F}(\text{or } \underline{F}) = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} - \frac{\partial U}{\partial z}\hat{k}$

Or short-hand it:  $\vec{F}(\text{or } \underline{F}) = -\nabla U$  (eq. 4.33)

Let's discuss this del/nabla ( $\nabla$ ) operator later.

# Non-linear potential field example:

$$U = \frac{x^2}{2} - \frac{x^3}{3}$$



1.  $\underline{f} = -\frac{\partial U}{\partial x} = ?$

2. Guess fixed points  $\bar{x}$  (where  $\underline{f}(\bar{x}) = 0$ ):

3. Linearization:

A good approxi around  $\bar{x}$  with a small  $\Delta x$ :  $\underline{f}(x) = \frac{D\underline{f}}{Dx}(\bar{x})\Delta x$

4. Calc. “eigenstuff”, then draw phase portrait and guess!

Def. Jacobian:

$$\frac{D\underline{f}}{Dx} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} \quad \text{When: } \underline{f} = \begin{bmatrix} f_1(x_1, x_2) \\ f_2(x_1, x_2) \end{bmatrix}$$



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# Non-linear potential field example, another

$$U = \frac{x^4}{4} - \frac{x^2}{2}$$



1.  $\underline{f} = -\frac{\partial U}{\partial x} = ?$

2. Guess fixed points  $\bar{x}$  (where  $\underline{f}(\bar{x}) = 0$ ):

3. Linearization:

A good approxi around  $\bar{x}$  with a small  $\Delta x$ :  $\underline{f}(x) = \frac{D\underline{f}}{Dx}(\bar{x})\Delta x$

4. Calc. “eigenstuff”, then draw phase portrait and guess!

Def. Jacobian:

$$\frac{D\underline{f}}{Dx} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} \quad \text{When: } \underline{f} = \begin{bmatrix} f_1(x_1, x_2) \\ f_2(x_1, x_2) \end{bmatrix}$$



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# Classic Mechs EP5.2 The del/nabla ( $\nabla$ ) operator

- Gradient:  $\text{grad } f = \nabla f$

$$f : \text{scalar} \longrightarrow \nabla f : \text{vector}$$

- Divergence:  $\text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$

$$\mathbf{F} : \text{vector} \longrightarrow \nabla \cdot \mathbf{F} : \text{scalar}$$

- Curl:  $\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$

$$\mathbf{F} : \text{vector} \longrightarrow \nabla \times \mathbf{F} : \text{vector}$$

# The del/nabla ( $\nabla$ ) operator

<https://en.wikipedia.org/wiki/Del>

- Gradient:  $\text{grad } f = \nabla f$

$$\nabla f = \left( \mathbf{e}_x \frac{\partial}{\partial x} \right) f + \left( \mathbf{e}_y \frac{\partial}{\partial y} \right) f + \left( \mathbf{e}_z \frac{\partial}{\partial z} \right) f = \frac{\partial f}{\partial x} \mathbf{e}_x + \frac{\partial f}{\partial y} \mathbf{e}_y + \frac{\partial f}{\partial z} \mathbf{e}_z$$

- Divergence:  $\text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$

$$\nabla \cdot \mathbf{F} = \left( \mathbf{e}_x \frac{\partial}{\partial x} \cdot \mathbf{F} \right) + \left( \mathbf{e}_y \frac{\partial}{\partial y} \cdot \mathbf{F} \right) + \left( \mathbf{e}_z \frac{\partial}{\partial z} \cdot \mathbf{F} \right) = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

- Curl:  $\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$

$$\nabla \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix}$$

$$\begin{aligned} \nabla \times \mathbf{F} &= \left( \mathbf{e}_x \frac{\partial}{\partial x} \times \mathbf{F} \right) + \left( \mathbf{e}_y \frac{\partial}{\partial y} \times \mathbf{F} \right) + \left( \mathbf{e}_z \frac{\partial}{\partial z} \times \mathbf{F} \right) \\ &= \frac{\partial}{\partial x} (0, -F_z, F_y) + \frac{\partial}{\partial y} (F_z, 0, -F_x) + \frac{\partial}{\partial z} (-F_y, F_x, 0) \end{aligned}$$

$$= \left( \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \mathbf{e}_x + \left( \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) \mathbf{e}_y + \left( \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \mathbf{e}_z$$